

Summability degree is not a homotopy invariant

A counterexample to the question in arXiv:1401.2123

Autonomous proof packet, lane 14

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Status. candidate_counterexample_likely_valid.

1 Source question

The source is M. Goffeng and B. Mesland, “Spectral triples and finite summability on Cuntz–Krieger algebras,” arXiv:1401.2123, published in *Documenta Mathematica* 20 (2015), 89–170 [1]. For $x \in K^i(A)$ they define $\text{sd}(x)$ to be the infimum of $p > 0$ for which x has a p -summable bounded Fredholm-module representative, and

$$\text{sd}_i(A) = \sup\{\text{sd}(x) : x \in K^i(A)\}.$$

On source PDF page 6 they ask whether this is a homotopy invariant.

Cuntz–Krieger algebras, which also are purely infinite in many cases. We furthermore provide a class of examples of θ -summable unbounded Fredholm modules on Cuntz–Krieger algebras such that their phases are finitely summable. This difference in finite summability for bounded and unbounded Fredholm modules indicates not only that lifting is a delicate matter but that the same holds for finer analytic properties of bounded transforms and the choice of χ .

Definition 4. Let A be a C^* -algebra. We say that the class $x \in K^*(A)$ is *summable of degree p* if there exists a p -summable Fredholm module representing x . We define the degree of summability of x to be the infimum of the set of numbers $p > 0$ for which x is p -summable. The odd (even) degree of summability of A is the supremum of the degree of summability of all odd (even) K -homology classes.

When taking the infimum of a set, we always apply the convention that the infimum of the empty set is infinite. We say that the C^* -algebra A has *finitely summable odd respectively even K -homology* if it has a finite odd respectively even degree of summability. We say that a K -homology class is finitely summable if it has a finite degree of summability. The summability degree of a C^* -algebra is clearly an isomorphism invariant. An interesting question is if it is a homotopy invariant. A related open problem hinted above is to find obstructions for finite summability of K -homology classes similar to the tracial obstructions for finitely summable unbounded Fredholm modules from [15]. We also note the terminology *uniformly summable* from [28].

Definition 5 ([28]). A C^* -algebra A is said to be *uniformly summable* if there is a $p > 0$ and a dense $*$ -subalgebra $\mathcal{A} \subseteq A$ such that any $x \in K^*(A)$ admits a representative that is p -summable on $\mathcal{A}$.

Example. We have one example of a K -homology class that is not finitely summable. This result is not to be confused with the interesting results of [57, 58] where a subalgebra of A , on which K -homology classes are required to be finitely summable, is fixed.

Lemma 6. Let $A := \bigoplus_{j=1}^{\infty} C(S^{2j-1})$ – the C_0 -direct sum of odd-dimensional spheres. There is a K -homology class $x \in K^1(A)$ with infinite degree of summability.

Proof. Consider the sum of fundamental classes $x = \sum_{i=1}^{\infty} [S^{2j-1}]$, that is, x is represented by

We answer this negatively already in odd degree, with homotopy-equivalent separable C^* -algebras whose degrees are ∞ and 0.

2 Proof intuition

The source paper supplies both extremes needed for a counterexample: a c_0 -sum of growing odd spheres carrying a class of infinite degree, and Cuntz–Krieger algebras whose every odd class has degree zero. The missing ingredient is to make those algebras homotopy equivalent without losing the analytic contrast.

The mapping cylinder does exactly this. Given $f : \mathcal{A} \rightarrow \mathcal{B}$, its mapping cylinder M_f is homotopy equivalent to $\mathcal{A}$, but evaluation at the free endpoint is a surjection $M_f \twoheadrightarrow \mathcal{B}$. Summability pulls back along surjections: the inverse image of a dense smooth subalgebra is dense. We choose $\mathcal{B}$ to be a c_0 -sum of one rapidly summable Cuntz–Krieger algebra and choose f to induce an isomorphism on odd K-homology. Then every class on M_f comes from $\mathcal{B}$ and has degree zero, although M_f is homotopy equivalent to the infinite-degree sphere sum.

3 Two elementary lemmas

We use the source convention for p -summable bounded Fredholm modules. All cycles used below may be chosen normalized, with $F = F^*$ and $F^2 = 1$, so only the commutator condition is visible.

Lemma 1 (Surjective pullback). *Let $q : C \twoheadrightarrow D$ be a surjective $*$ -homomorphism. If $y \in K^i(D)$ has a p -summable representative, then $q^*y \in K^i(C)$ has a p -summable representative.*

Proof. Let (π, H, F) represent y , and let $D_p \subseteq D$ be a dense $*$ -subalgebra on which the Schatten conditions hold. The pulled-back cycle is $(\pi q, H, F)$, and the same conditions hold on $q^{-1}(D_p)$.

It remains only to check density. Given $c \in C$, choose $d_n \in D_p$ with $d_n \rightarrow q(c)$. Since a surjective bounded linear map between Banach spaces is open, there are $u_n \in C$ with

$$q(u_n) = d_n - q(c), \quad \|u_n\| \leq 2\|d_n - q(c)\|.$$

Then $c + u_n \rightarrow c$ and $q(c + u_n) = d_n \in D_p$. Thus $q^{-1}(D_p)$ is dense. $\square$

Lemma 2 (A c_0 -sum of degree-zero targets). *Let D be a C^* -algebra such that every odd K-homology class has a normalized p -summable representative for every $p > 0$. Then*

$$\mathcal{B} := \bigoplus_{j=1}^{\infty} D$$

has $\text{sd}_1(\mathcal{B}) = 0$.

Proof. By countable additivity of KK-theory in the first variable, $K^1(\mathcal{B}) \cong \prod_j K^1(D)$. Fix $y = (y_j)$ and $p > 0$. Choose a normalized p -summable cycle (π_j, H_j, F_j) for each y_j , and take their Hilbert direct sum. For $b = (b_j) \in \mathcal{B}$,

$$[F, \pi(b)] = \bigoplus_j [F_j, \pi_j(b_j)].$$

Every block is compact and its norm is at most $2\|b_j\| \rightarrow 0$, so this direct sum is compact. The algebraic direct sum of the coordinate smooth subalgebras is a dense $*$ -subalgebra of $\mathcal{B}$; on it the commutator is a finite direct sum of Schatten- p operators. The defects vanish because the cycles are normalized. Thus y is p -summable. Since $p > 0$ and y were arbitrary, $\text{sd}_1(\mathcal{B}) = 0$. $\square$

4 The Cuntz–Krieger target and the KK-equivalences

Consider the 0–1 matrix

$$Q = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}. \quad (1)$$

Its directed graph is strongly connected and is not a permutation graph, so $D := O_Q$ is a unital simple purely infinite Cuntz–Krieger algebra [4]. Moreover,

$$I - Q^\top = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

The relations given by the last two columns kill the first coordinate and identify the third generator with the negative of the second. Thus the cokernel is $\mathbb{Z}$. The kernel is generated by $(0, 1, -1)$, and hence

$$K_0(D) = \text{coker}(I - Q^\top) \cong \mathbb{Z}, \quad K_1(D) = \ker(I - Q^\top) \cong \mathbb{Z}. \quad (2)$$

Equivalently, the Smith invariants are $(1, 1, 0)$.

Goffeng–Mesland’s Theorem 7 says that $\text{sd}_1(O_Q) = 0$ and, more precisely, constructs normalized p -summable analytic K-cycles for every $p > 0$ and every odd class [1]. Thus D satisfies Lemma 2.

summability of any $z \in K^1(C(S^{2k-1})) \setminus \{0\}$ given by $2k - 2$. Hence we have a contradiction since our assumptions imply that $p \geq 2k - 2$ for all k . $\square$

Content and organization of the paper. The content of the paper can be summarized in the following Theorem. We use the letter A for an $N \times N$ -matrix only containing the numbers 0 and 1. It will be clear from its context when A is a matrix and when it is a C^* -algebra as above.

The notation Ω_A is for the space of characters of the standard maximal abelian subalgebra in the Cuntz-Krieger algebra O_A associated with the $N \times N$ -matrix A , see more in Section 4 below. There is a $C(\Omega_A)$ -valued conditional expectation on O_A . We let $\mathcal{E}_A^\Omega$ denote the $C(\Omega_A)$ -Hilbert C^* -module closure of O_A and $\pi_A^\Omega : O_A \rightarrow \text{End}_{C(\Omega_A)}^*(\mathcal{E}_A^\Omega)$ the left O_A -action.

Theorem 7. *The odd degree of summability of a Cuntz Krieger algebra O_A is 0. Moreover, there exists an unbounded bivariate $(O_A, C(\Omega_A))$ -cycle $(\mathcal{E}_A^\Omega, D)$ with the following property. For each class $x \in K^1(O_A)$ one can find a finite collection $(\omega_{j,x})_{j=1}^{m_x} \subseteq \Omega_A$ and localize D at each of these characters to construct the self-adjoint operator $D_{j,x}$ on $\mathcal{E}_A^\Omega \otimes_{\omega_{j,x}} \mathbb{C}$ such that the unbounded Fredholm module*

$$\left(\bigoplus_{j=1}^{m_x} (\pi_A^\Omega \otimes_{\omega_{j,x}} \text{id}_{\mathbb{C}}), \bigoplus_{j=1}^{m_x} (\mathcal{E}_A^\Omega \otimes_{\omega_{j,x}} \mathbb{C}), \bigoplus_{j=1}^{m_x} D_{j,x} \right),$$

is a θ -summable representative for $x \in K^1(O_A)$. Moreover, each of the triples

$$\left(\pi_A^\Omega \otimes_{\omega_{j,x}} \text{id}_{\mathbb{C}}, \mathcal{E}_A^\Omega \otimes_{\omega_{j,x}} \mathbb{C}, D_{j,x} |D_{j,x}|^{-1} \right)$$

form p -summable analytic K-cycles for any $p > 0$.

Remark 8. The first statement of this Theorem should be compared to the results of [28]. The intersection of applications for this paper with [28] lies in the examples of discrete hyperbolic

Put $A_j = C(S^{2j-1})$. Both A_j and D belong to the UCT class and have free graded K-groups $(\mathbb{Z}, \mathbb{Z})$. The UCT therefore lifts any graded K-theory isomorphism to an invertible element

$$\kappa_j \in KK(A_j, D)[3].$$

Kirchberg's homomorphism realization theorem, in the form of Dadarlat's Theorem 2.3(i), applies because A_j is separable, nuclear, exact, and unital and D is separable, unital, and properly infinite. It gives a nuclear $*$ -homomorphism

$$f_j : A_j \longrightarrow D, \quad KK(f_j) = \kappa_j[2]. \quad (3)$$

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full $\mathcal{O}_2$ -factorable $*$ -homomorphism from A to B . Indeed, any separable exact C^* -algebra embeds in $\mathcal{O}_2$ by a theorem of Kirchberg [18, Thm. 6.3.11] and $\mathcal{O}_2$ admits a full embedding in B by [18, 4.1.4 and 4.2.3], since B has a full properly infinite projection. We shall need the following results of Kirchberg and Phillips, (see [18, Thm. 8.3.3] and [16, Thm. 4.1.1]). In order to ease notation, we will write $KK(\varphi)$ for the class of $*$ -homomorphism $\varphi : A \rightarrow B$ in the group $KK_{\text{nuc}}(A, B)$ of [21].

Theorem 2.3 (Kirchberg's Classification Theorem). *Let A be a separable exact unital C^* -algebra and let B be a separable unital properly infinite C^* -algebra. Fix a full $\mathcal{O}_2$ -factorable $*$ -homomorphism $\theta : A \rightarrow B$.*

(i) *For every $\alpha \in KK_{\text{nuc}}(A, B)$ there is a full nuclear $*$ -homomorphism $\varphi : A \rightarrow B$ such that $KK(\varphi) = \alpha$.*

(ii) *Let $\varphi, \psi : A \rightarrow B$ be two unital nuclear $*$ -homomorphisms. Then $\varphi \oplus \theta \approx_{\text{uh}} \psi \oplus \theta$ if and only if $KK(\varphi) = KK(\psi)$.*

(iii) *Assume that B is purely infinite and simple and let $\varphi, \psi : A \rightarrow B$ be two unital nuclear injective $*$ -homomorphisms. Then $\varphi \approx_{\text{uh}} \psi$ if and only if $KK(\varphi) = KK(\psi)$.*

Theorem 2.4 (Phillips' Classification Theorem). *Let A be a separable nuclear simple unital C^* -algebra and let B be a separable unital C^* -algebra that satisfies $B \cong B \otimes \mathcal{O}_\infty$.*

(i) *For every $\alpha \in KK(A, B)$ there is a full $*$ -homomorphism $\varphi : A \rightarrow B \otimes \mathcal{K}$*

Now set

$$\mathcal{A} = \bigoplus_{j=1}^{\infty} A_j, \quad \mathcal{B} = \bigoplus_{j=1}^{\infty} D, \quad f = \bigoplus_{j=1}^{\infty} f_j : \mathcal{A} \rightarrow \mathcal{B}. \quad (4)$$

The map is well-defined because each f_j is contractive. Countable additivity gives

$$K^1(\mathcal{B}) \cong \prod_j K^1(D), \quad K^1(\mathcal{A}) \cong \prod_j K^1(A_j),$$

and f^* is the product of the isomorphisms f_j^* . Hence

$$f^* : K^1(\mathcal{B}) \xrightarrow{\cong} K^1(\mathcal{A}). \quad (5)$$

5 The counterexample

Theorem 3. *The odd degree of summability is not invariant under C^* -homotopy equivalence. More precisely, there are separable homotopy-equivalent C^* -algebras $\mathcal{A}$ and M_f such that*

$$\text{sd}_1(\mathcal{A}) = \infty, \quad \text{sd}_1(M_f) = 0.$$

Proof. Use $\mathcal{A}, \mathcal{B}, f$ from (4), and form the mapping cylinder

$$M_f = \{(a, h) \in \mathcal{A} \oplus C([0, 1], \mathcal{B}) : h(0) = f(a)\}.$$

Let

$$q : M_f \rightarrow \mathcal{A}, \quad q(a, h) = a, \quad s : \mathcal{A} \rightarrow M_f, \quad s(a) = (a, \text{constant path } f(a)).$$

Then $qs = \text{id}_{\mathcal{A}}$. For $0 \leq t \leq 1$, define

$$H_t(a, h) = (a, h_t), \quad h_t(r) = h(tr).$$

Each H_t is a $*$ -homomorphism, the path is point-norm continuous, $H_0 = sq$, and $H_1 = \text{id}_{M_f}$. Thus q is a homotopy equivalence with inverse s .

Let

$$e : M_f \longrightarrow \mathcal{B}, \quad e(a, h) = h(1).$$

This is surjective: for $b \in \mathcal{B}$, the element $(0, h)$ with $h(r) = rb$ maps to b . Also $es = f$. Since $s^* : K^1(M_f) \rightarrow K^1(\mathcal{A})$ is an isomorphism, (5) implies that $e^* : K^1(\mathcal{B}) \rightarrow K^1(M_f)$ is surjective. Indeed, for $z \in K^1(M_f)$ choose $y \in K^1(\mathcal{B})$ with $f^*y = s^*z$; then

$$s^*(e^*y) = (es)^*y = f^*y = s^*z,$$

and injectivity of s^* gives $e^*y = z$.

By Lemma 2, every $y \in K^1(\mathcal{B})$ is p -summable for every $p > 0$. Lemma 1, applied to the surjection e , shows that every $z \in K^1(M_f)$ is p -summable for every $p > 0$. Hence $\text{sd}_1(M_f) = 0$.

On the other hand, $\mathcal{A}$ is exactly the c_0 -sum used in Goffeng–Mesland’s Lemma 6. The product class whose j th coordinate is the fundamental class of S^{2j-1} has infinite degree of summability [1]. Therefore $\text{sd}_1(\mathcal{A}) = \infty$. Since M_f and $\mathcal{A}$ are homotopy equivalent, this is the required counterexample. $\square$

6 Scope, verification, and novelty

The theorem gives a full negative answer in odd degree; one parity is enough to disprove homotopy invariance. It does not assert an even counterexample and does not solve the source’s separate broad problem of classifying all obstructions to finite summability.

The matrix calculation was checked both by hand and by the packet’s exact integer verifier. The proof was audited for nonunital c_0 -sums, Schatten exponents $0 < p < 1$, compactness of infinite block diagonals, contravariance of K-homology, and point-norm continuity of the cylinder contraction. A bounded exact-phrase and mapping-cylinder literature search found later repetitions of the question but no prior solution. Exhaustive priority is not claimed. Human review should concentrate on (3), Lemma 2, and the equality $es = f$ in the final K-homology diagram.

References

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- [2] M. Dadarlat, *The homotopy groups of the automorphism group of Kirchberg algebras*, Journal of Noncommutative Geometry 1 (2007), 113–139.
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